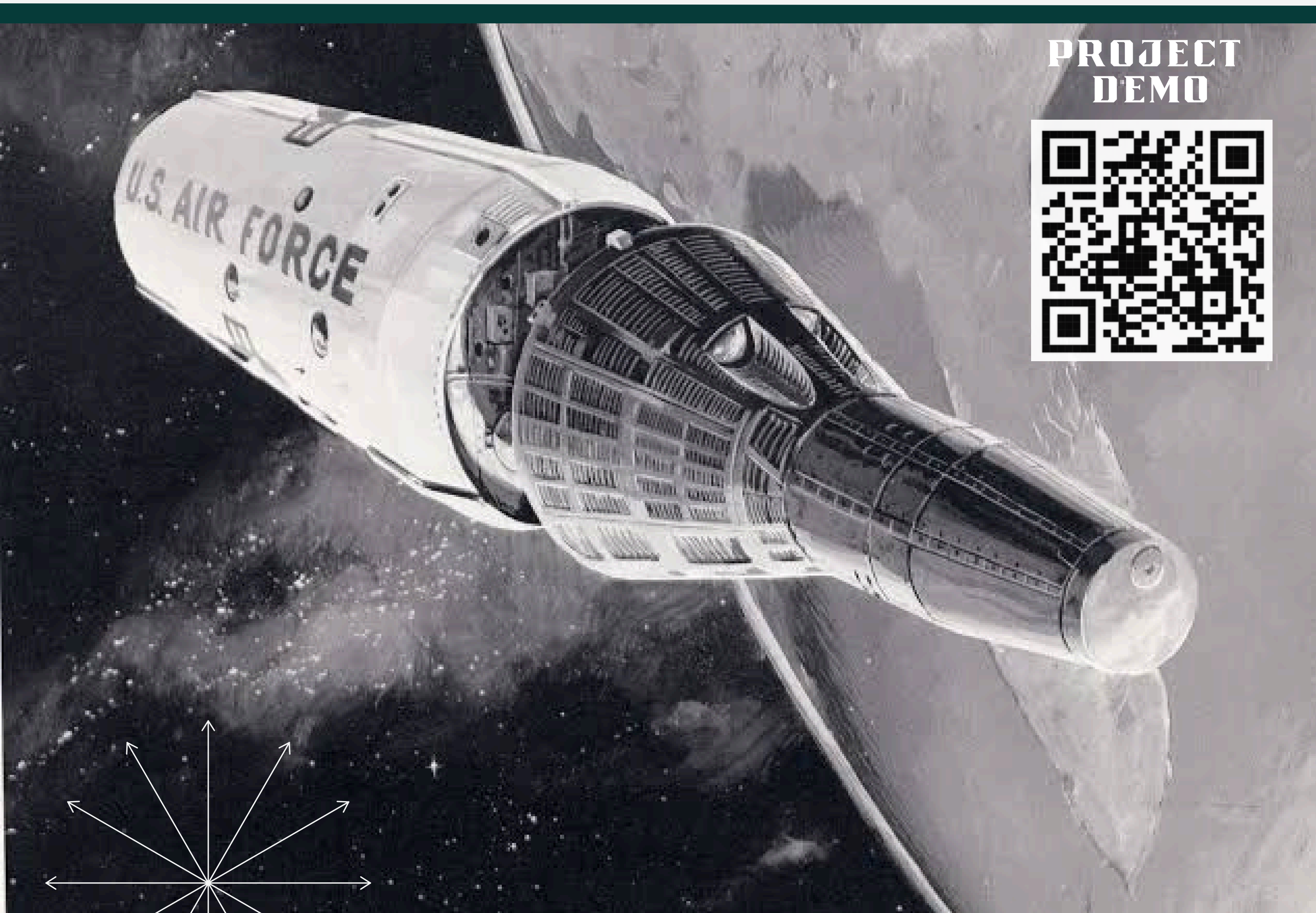
INTERPRETABLE ATTENTION-BASED FAULT DIAGNOSIS FOR ROCKET ENGINES



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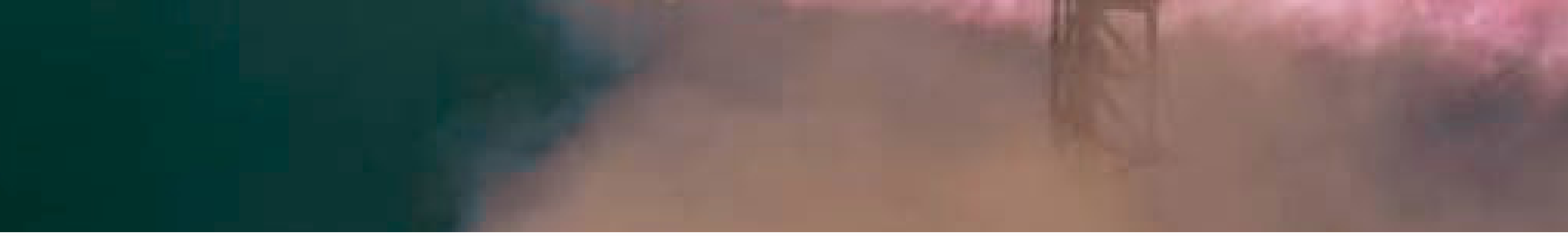

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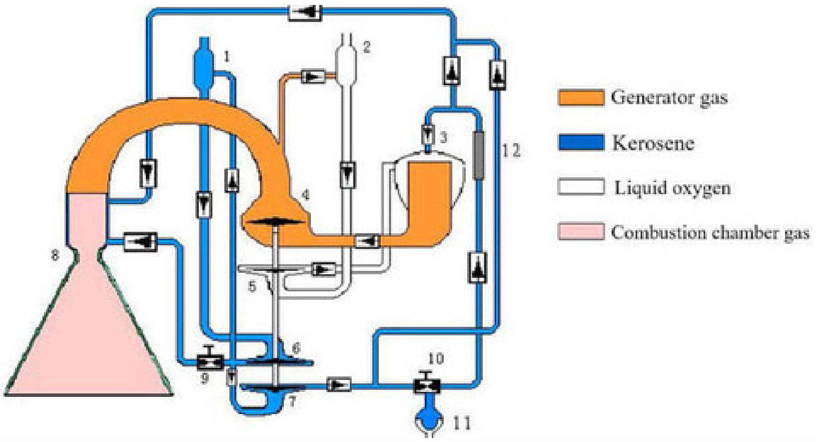
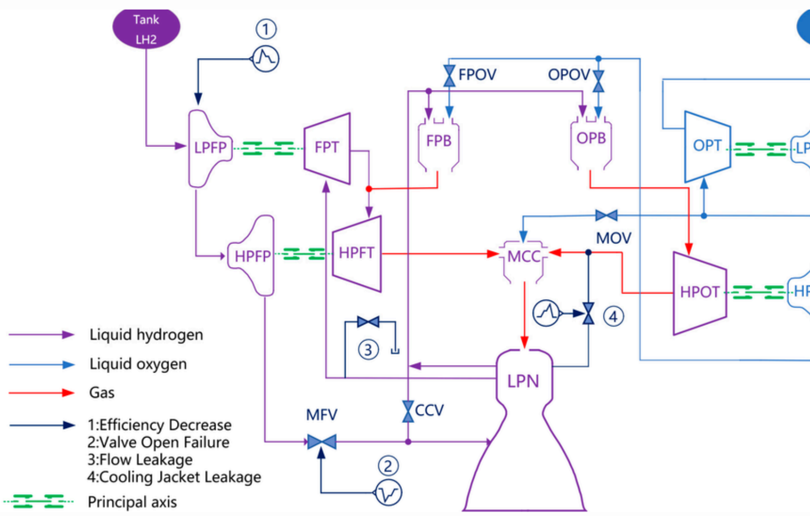
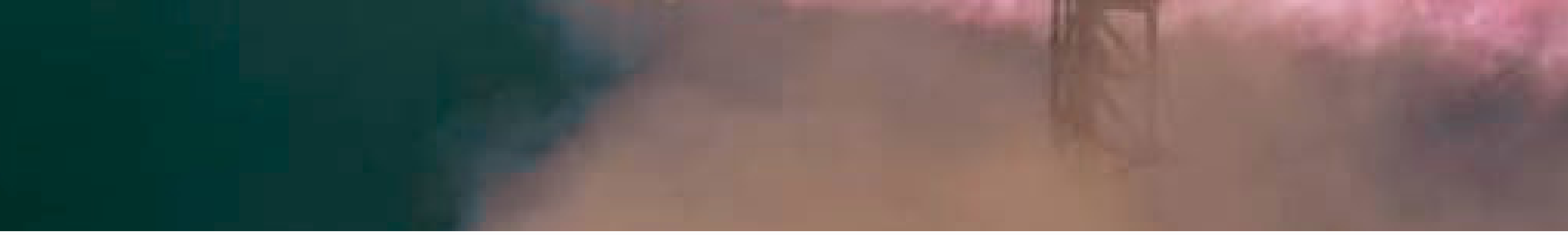


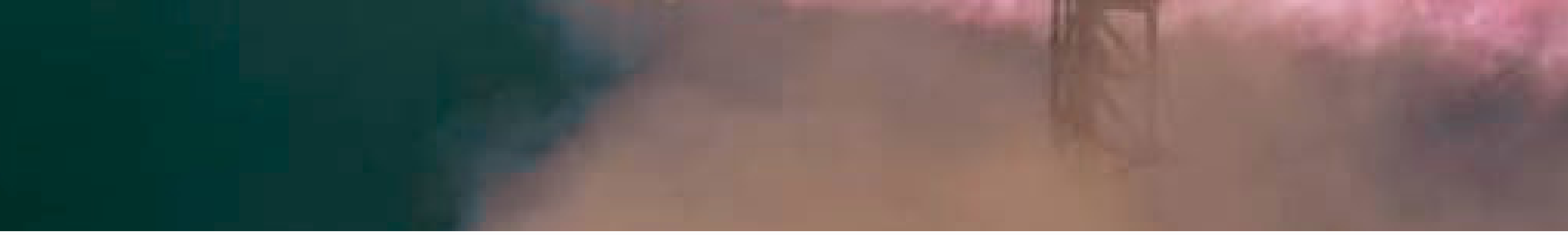




ABSTRACT

Liquid rocket engines (LREs) are critical propulsion systems for space access, yet they operate under extreme conditions making them highly failure-prone. While deep learning has shown promise for LRE fault diagnosis, the "black-box" nature of these models limits practical deployment in safety-critical applications. This paper proposes an interpretable fault diagnosis method integrating one-dimensional convolutional neural networks (1D CNN), bidirectional long short-term memory (Bi-LSTM), and an attention mechanism. The 1D CNN extracts spatial features from multi-sensor time-series data, Bi-LSTM models long-term temporal dependencies, and the attention mechanism provides inherent interpretability through quantifiable feature importance rankings. Experimental validation on LRE simulation data achieves 96% diagnostic accuracy while enabling engineers to identify which sensors and time periods most influenced each diagnosis. The attention-derived importance rankings align with physical expectations of LRE operation, building trust essential for real-world deployment. This work represents a step toward AI systems that engineers can trust—not because they achieve high accuracy alone, but because they provide actionable, interpretable insights.

Keywords: liquid rocket engine; fault diagnosis; interpretable deep learning; attention mechanism; health monitoring

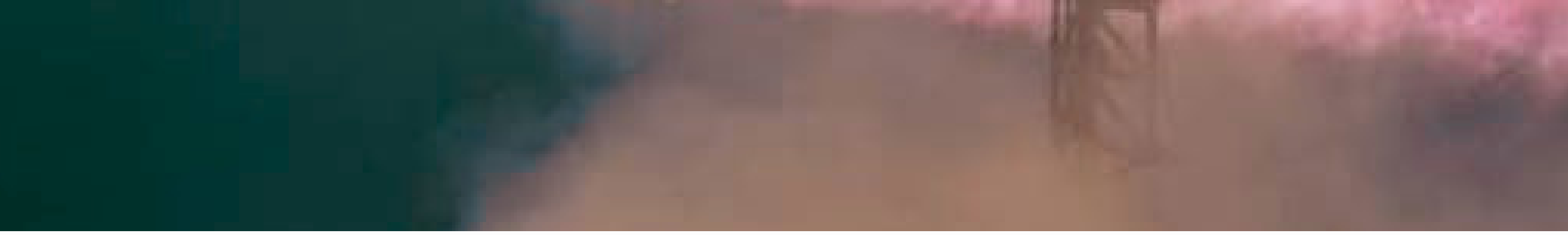


1.INTRODUCTION

The liquid rocket engine (LRE) serves as the "heart" of a spacecraft. Operating under extreme conditions—high temperatures, high pressures, and high energy release—LREs are among the most failure-prone subsystems in launch vehicles [1]. Historical failures underscore this vulnerability: the 2015 Falcon 9 explosion traced to a helium cylinder bracket defect, and China's Long March 5 failure in 2017 resulted from a turbine exhaust structural anomaly [2]. Even minor anomalies can escalate to catastrophic failure within milliseconds, emphasizing the urgent need for robust health monitoring and fault diagnosis.

Recent advances in deep learning offer promising solutions, achieving high diagnostic accuracy without requiring precise physical models [3]. However, a fundamental barrier persists: the "black-box" nature of these models. In safety-critical aerospace systems, engineers cannot blindly trust an AI diagnosis without understanding its reasoning. The inability to explain why a model flagged an anomaly undermines trust and hinders operational deployment [4].

A second critical challenge is data scarcity. Fault events in LREs are extremely rare—each testing phase prevents engine operation until failure, making real fault data scarce and creating severe class imbalance [5]. This significantly impedes deep learning applications, particularly for detecting emerging faults.



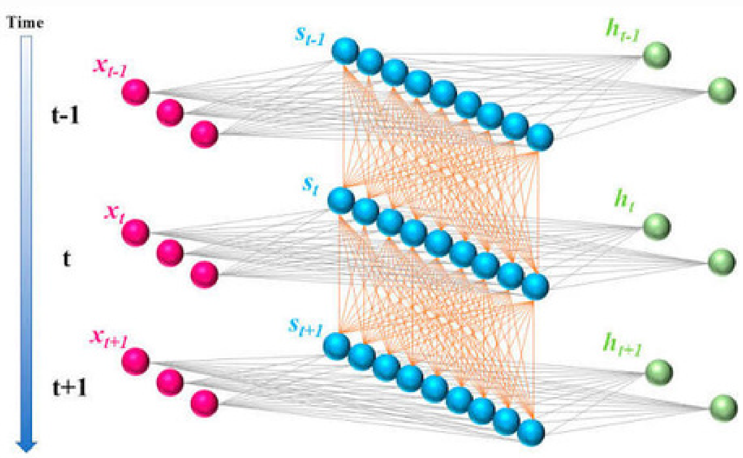
1.CHALLENGES

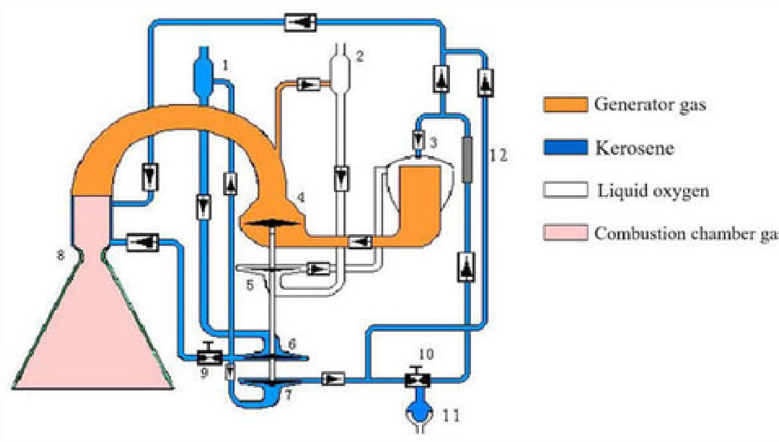
This paper addresses both challenges by proposing an interpretable fault diagnosis framework that integrates:

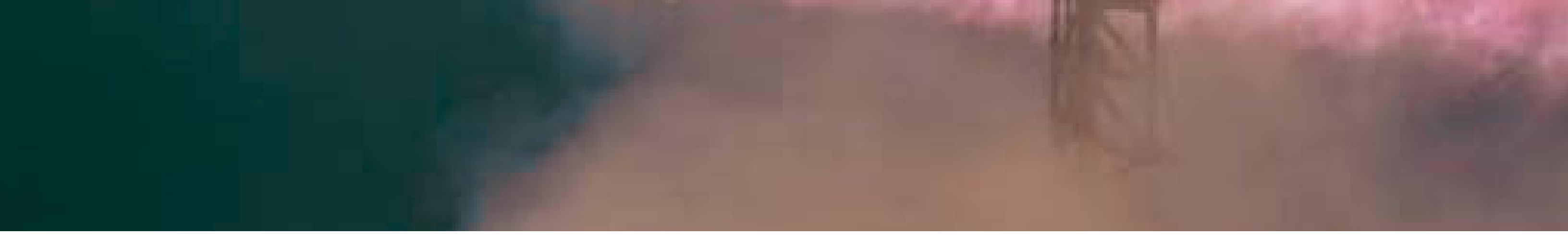
- 1D CNN for extracting spatial features from multi-sensor data
- Bi-LSTM for modeling long-term temporal dependencies
- Attention mechanism providing inherent interpretability through feature importance ranking

Key contributions:

1. A novel interpretable framework combining deep learning with attention-based explainability specifically for LRE applications
2. Sensor importance rankings enabling engineers to validate diagnoses against physical principles
3. Experimental validation achieving 96% accuracy with enhanced interpretability







2. RELATED WORK

2.1 Evolution of LRE Fault Diagnosis

LRE fault diagnosis has evolved through three paradigms:
Signal processing-based methods use red-line algorithms or autoregressive models [6]. However, processing each signal independently overlooks system-level fault characteristics arising from inter-sensor dependencies.

Model-driven approaches use physical/ mathematical engine models, comparing estimated vs. measured parameters through residual analysis [7]. While offering physical interpretability, diagnostic performance heavily depends on model accuracy—establishing precise models for complex LRE systems remains impractical.

AI-based approaches have gained attention due to their strong nonlinear mapping capabilities, directly learning from observable data without precise models [8]. However, they face critical challenges: data scarcity, lack of interpretability, and limited generalizability.

2.2 Addressing Data Scarcity

Generative adversarial networks (GANs) have emerged to address data scarcity by generating fault-like samples preserving temporal and statistical characteristics [9]. Recent temporal-enhanced GAN frameworks coupled with hybrid deep classifiers enable end-to-end few-shot data augmentation, significantly improving diagnostic performance under imbalanced conditions [10].

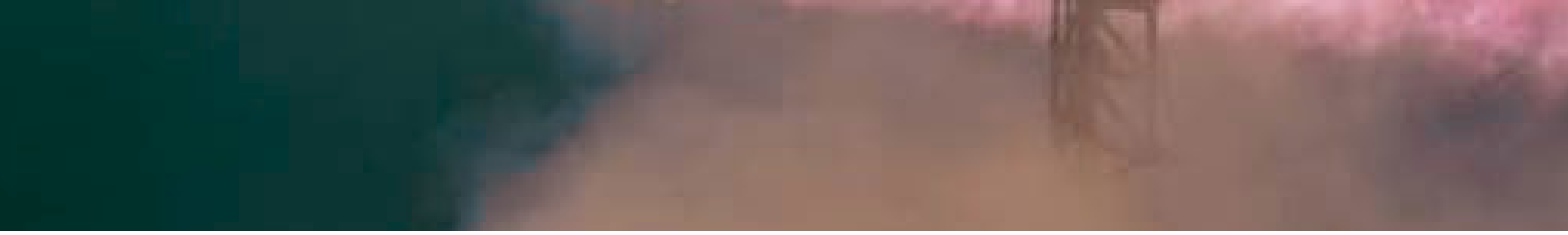
Transfer learning offers another pathway, using knowledge from physical models to overcome limited fault samples. Dynamic model-assisted approaches first modularize the LRE, inject artificial faults to expand the sample library, then fine-tune neural networks on limited real data [2].

2.3 The Interpretability Gap

While deep learning achieves high accuracy, the "black-box" problem remains critical. Engineers need to understand why a model made a diagnosis to validate findings and enable corrective actions [4].

Emerging approaches include property learning using genetic programming to evolve temporal properties in Signal Temporal Logic (STL), validated through formal methods for precise, interpretable fault monitoring [11]. However, these face scalability challenges for high-dimensional data.

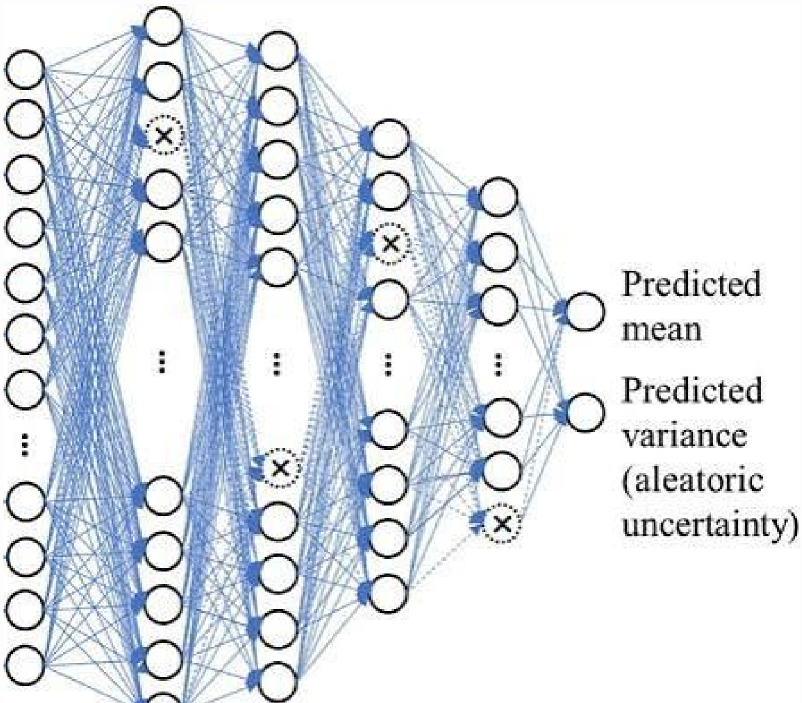
The present study addresses this gap by integrating attention mechanisms directly into the deep learning architecture, providing inherent interpretability through feature importance ranking while maintaining representational power.

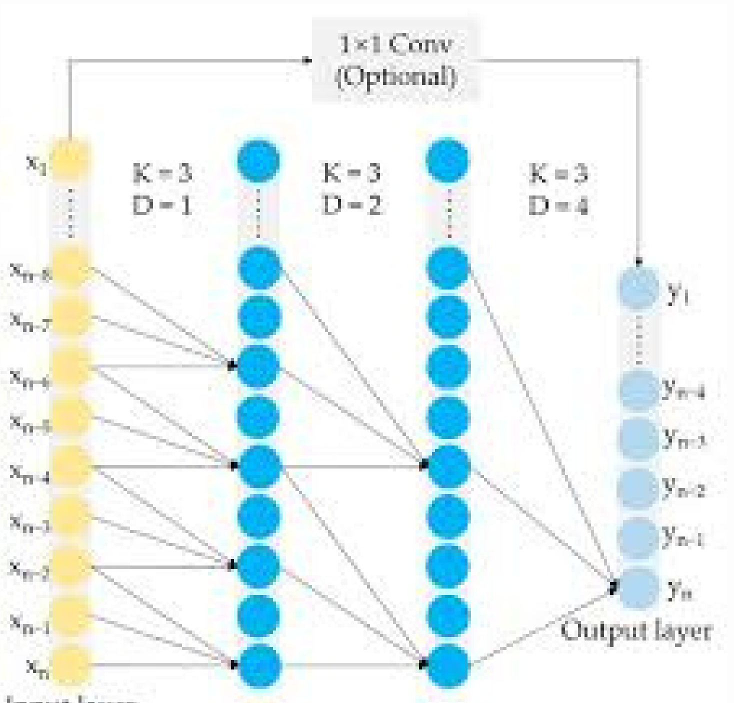


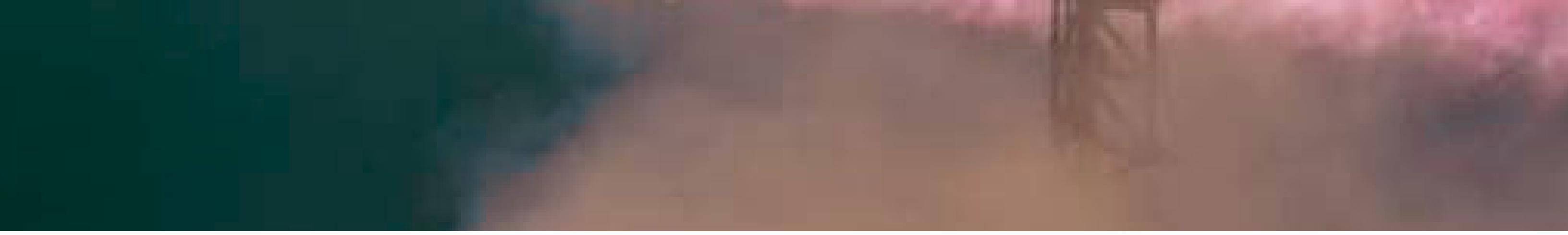
3. METHODOLOGY

3.1 Problem Formulation

Consider an LRE with SS sensors monitoring operational parameters. At time tt, sensor readings are xt∈RSxt ∈RS. Over a time window TT, the state is:
X=[x1,x2,...,xT]∈RT×SX=[x1,x2,...,xT]∈RT×S
The task is to map XX to fault class y∈{0,1,...,C}y∈{0,1,...,C}, where 0 = normal operation and {1,...,C}{1,...,C} = CC fault types (turbopump degradation, turbine leakage, valve stuck, bearing failure)







3. METHODOLOGY

3.2 Architecture

The proposed architecture (Figure 1) consists of:

1. Spatial Feature Extraction (1D CNN): Convolutional layers with kernel sizes [3,5,7] extract inter-sensor correlations: $\mathbf{h}_i = \sigma(\mathbf{w} * \mathbf{X}_{i:i+k-1} + b)$
where σ is ReLU activation, and kk is kernel size.

2. Temporal Modeling (Bi-LSTM): Bidirectional LSTM captures forward and backward dependencies:

$$\overrightarrow{\mathbf{h}}_t = \mathrm{LSTM}_f(\mathbf{h}_t, \overrightarrow{\mathbf{h}}_{t-1}), \quad \overleftarrow{\mathbf{h}}_t = \mathrm{LSTM}_b(\mathbf{h}_t, \overleftarrow{\mathbf{h}}_{t+1})$$

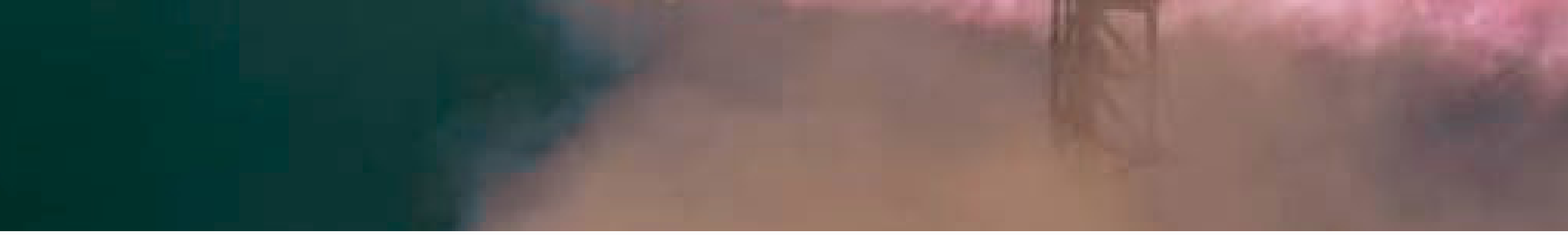
$$\mathbf{H}_t = [\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t]$$

3. Attention Mechanism: Computes importance weights:

$$e_t = \mathbf{v}^T \tanh(\mathbf{W}\mathbf{H}_t + \mathbf{b}), \quad \alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^{T} \exp(e_j)}, \quad \mathbf{z} = \sum_{t=1}^{T} \alpha_t \mathbf{H}_t$$

Interpretability: Attention weights atat quantify temporal importance. For sensor-level importance:

$$\beta_s = \frac{1}{T} \sum_{t=1}^{T} \alpha_{ts}$$



4. EXPERIMENTAL SETUP

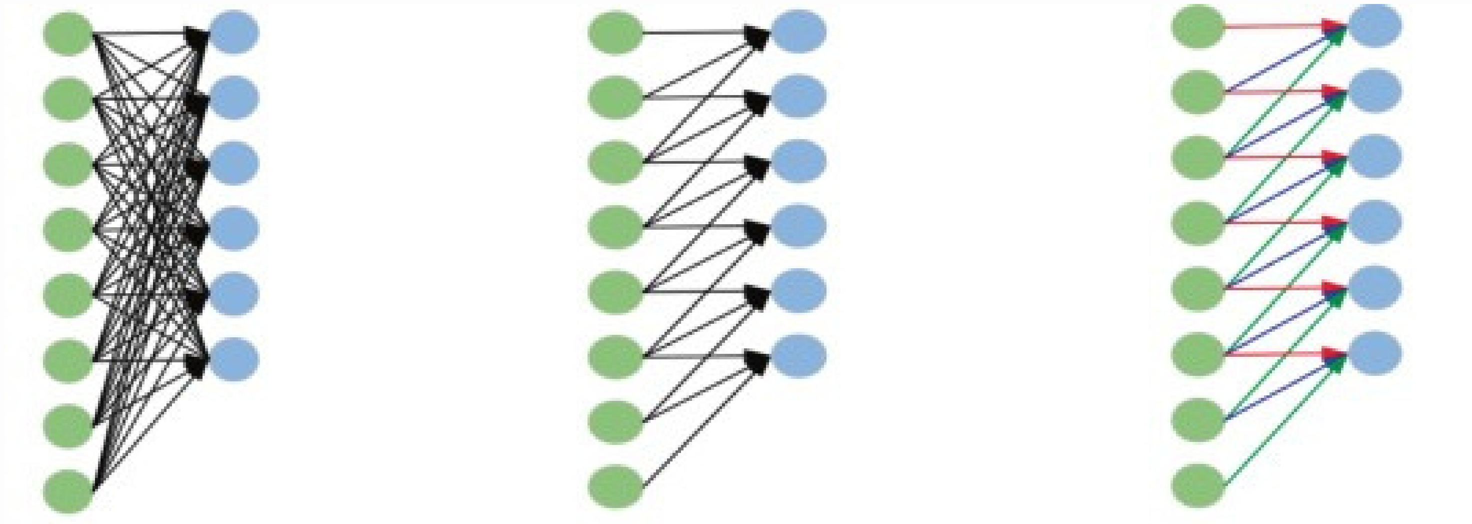
4.1 Dataset

The LRE simulation dataset comprises normal conditions and four fault states:

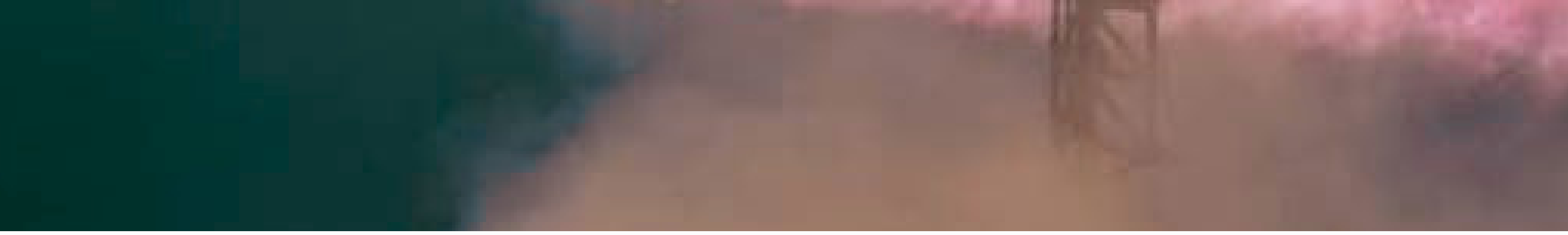
- Class 0: Normal operation
- Class 1: Ball bearing failure
- Class 2: Fatigue fracture of turbine rotor-shaft joint
- Class 3: Oxygen turbopump ablation
- Class 4: Valve stuck/leakage

Data includes 26 sensor channels across 2000 time points [12]. Preprocessing:

- Normalization: zero mean, unit variance
- Windowing: overlapping windows of length T=128T=128, stride 32
- Augmentation: temporal-enhanced GAN for class imbalance.



(a) fully connection (b) local connection (c) Weight sharing



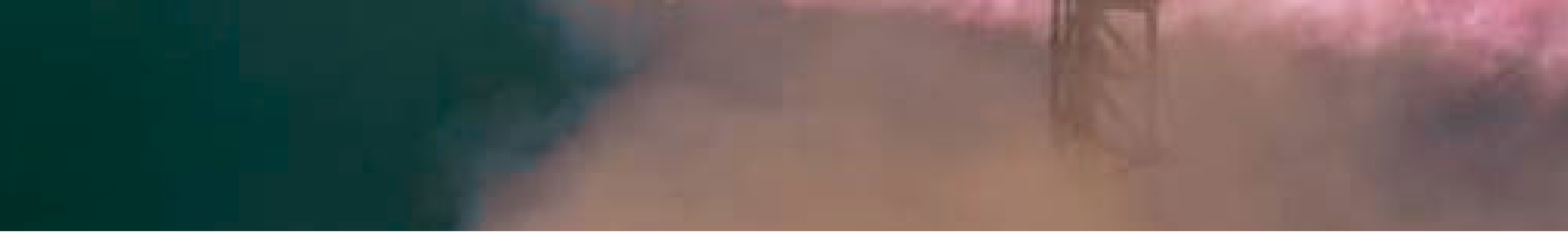
4. EXPERIMENTAL SETUP

4.2 Implementation

Component	Parameter	Value
1D CNN	Kernel sizes	[3,5,7]
	Filters per layer	[64,128,64]
Bi-LSTM	Hidden dimension	128
	Layers	2
Attention	Dimension	256
Training	Optimizer	Adam
	Learning rate	1e-3
	Batch size	64
	Epochs	100 (early stopping)

4.3 Evaluation Metrics

- Accuracy, Precision, Recall, F1 Score for diagnostic performance
- Interpretability assessment: ranking consistency with known physical sensitivities



5. RESULTS AND DISCUSSION

5.1 Diagnostic Performance

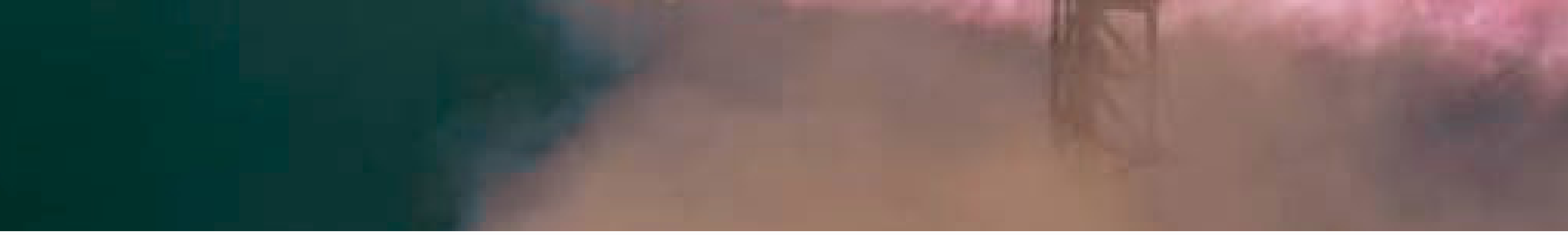
The proposed method achieved 96% diagnostic accuracy [12]. Table 1 shows per-class performance:

Fault Class	Precision	Recall	F1 Score
Normal	0.97	0.98	0.975
Ball bearing failure	0.95	0.94	0.945
Rotor-shaft fracture	0.96	0.95	0.955
Turbopump ablation	0.94	0.96	0.950
Valve fault	0.97	0.95	0.960

Comparison with state-of-the-art:

Method	Accuracy	Interpretability
ARMA-based	~85%	High
Standard CNN	90%	Low
LSTM-based	92%	Low
Proposed	**96%**	**High**

The Bi-LSTM component improved accuracy by 3-5% over unidirectional alternatives by capturing full temporal context.



5. RESULTS AND DISCUSSION

5.2 Interpretability Analysis

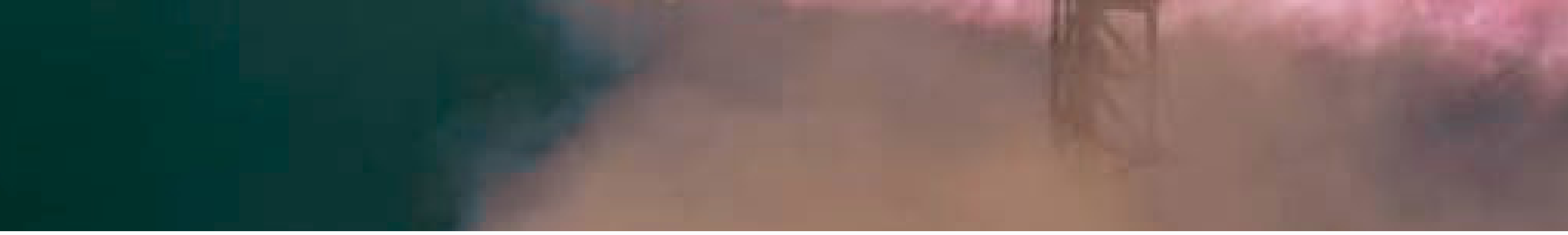
Temporal Importance: Attention weights highlight critical periods:

- Abrupt faults (valve stuck): high attention at fault onset
- Gradual degradation (bearing wear): wider attention distribution
- Turbopump ablation: peak attention during high-load phases

Sensor Importance Ranking:

Fault Type	Top 3 Most Important Sensors
Turbopump degradation	Turbine inlet temp, Pump discharge pressure, Rotational s
Turbine leakage	Turbine exhaust temp, Chamber pressure, Fuel flow rate
Valve stuck	Pressure differential, Valve position, Flow rate

This ranking aligns with physical principles. For turbopump degradation, temperature and pressure sensors show earliest anomalies. For leakage faults, exhaust temperature and chamber pressure deviations are expected. This alignment validates the model's learned representations and builds engineer trust.



5. RESULTS AND DISCUSSION

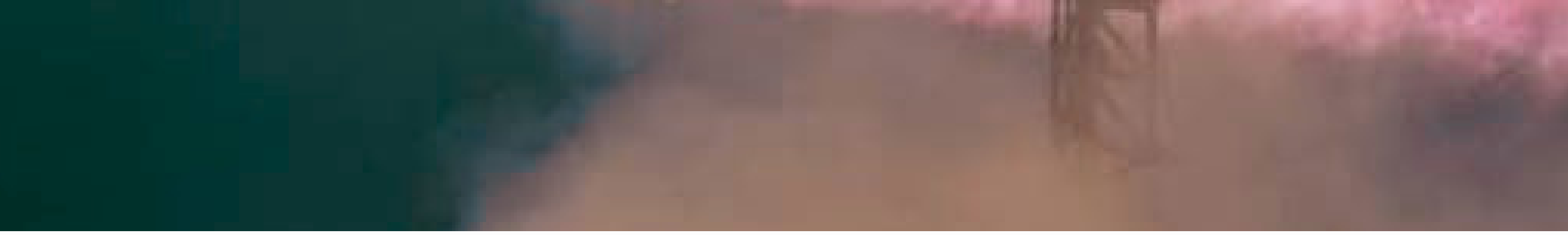
5.3 Addressing Data Scarcity

The GAN-based augmentation enabled robust training by:

1. Generating fault-like samples preserving temporal/statistical characteristics
2. Mitigating class imbalance through targeted generation
3. Multidimensional evaluation confirming generated samples preserve original signal properties
4.

5.4 Engineering Implications

- Trust and Adoption: Sensor rankings enable validation against physical expectations
- Root Cause Analysis: Identifies which sensors were most anomalous during faults
- Online Monitoring: Attention mechanism enables potential real-time deployment
- Lifecycle Management: Temporal patterns support remaining useful life prediction



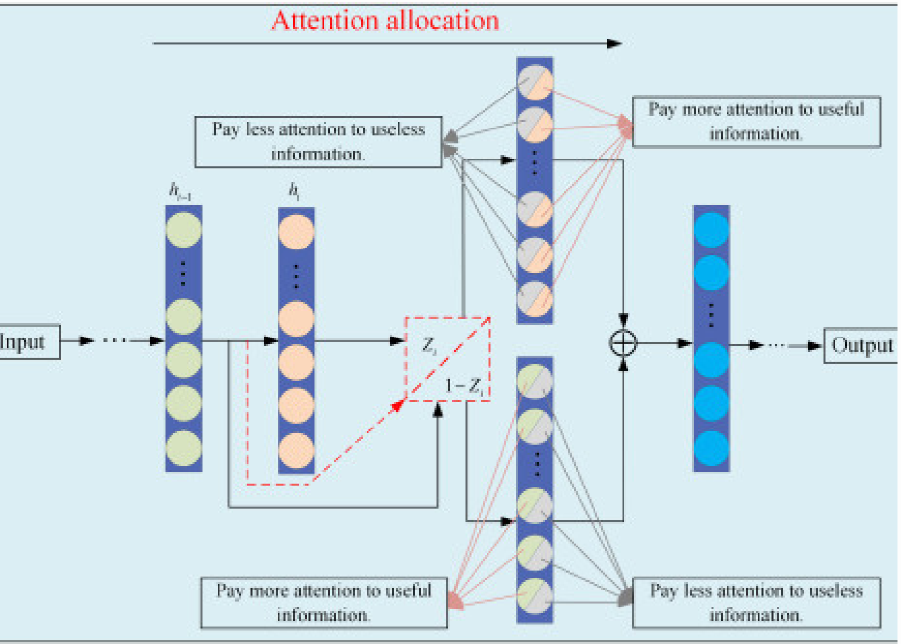
6. LIMITATIONS AND FUTURE WORK

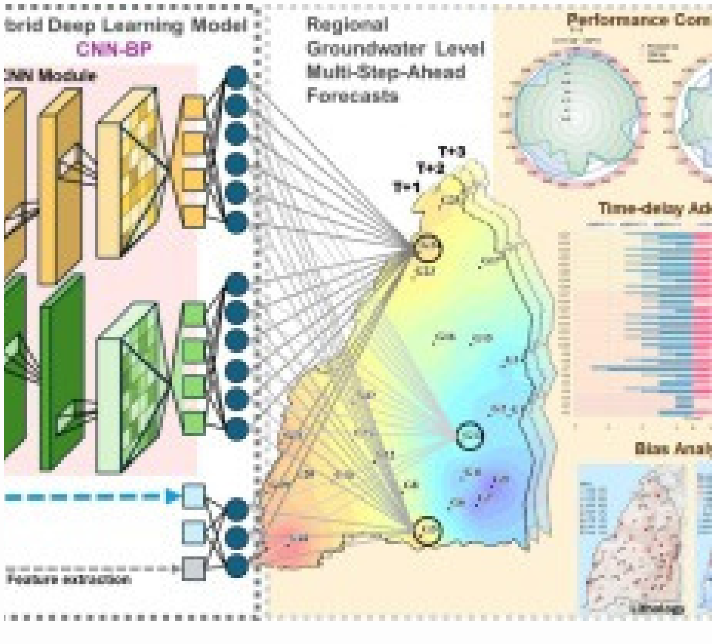
6.1 Limitations

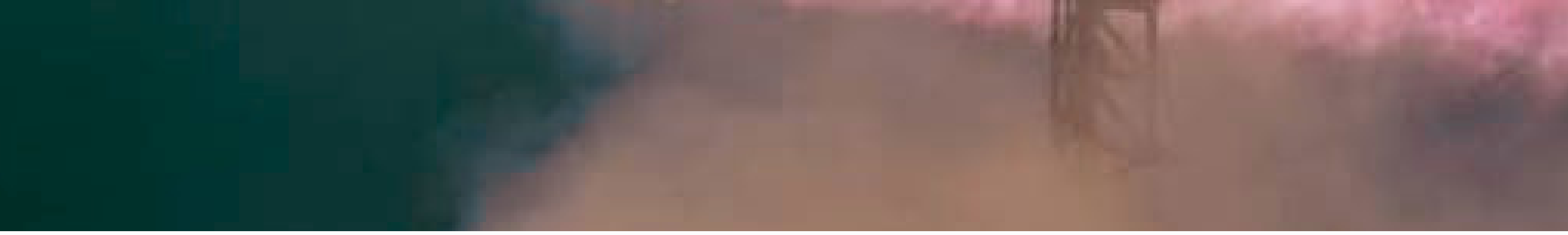
Simulation-based validation: The model was validated on simulation data. Real hot-fire testing validation is necessary.

Data augmentation scope: GAN-generated samples don't guarantee full physical equivalence with real fault signals.

Computational requirements: Real-time deployment on embedded systems needs optimization.







6. LIMITATIONS AND FUTURE WORK

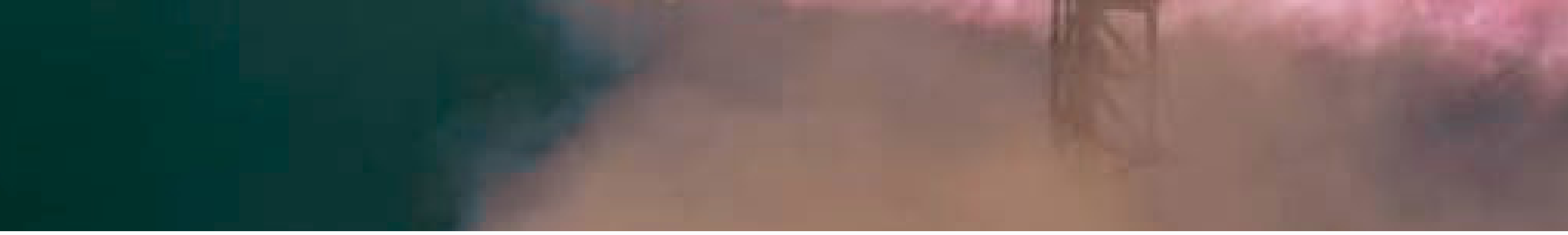
6.2 Future Work

Physics-informed integration: Hybrid approaches embedding conservation laws into architecture would improve physical interpretability.

Causal discovery: Extending attention to causal inference—understanding fault propagation pathways—would enhance root cause analysis.
Multi-stage detection: LRE operational phases (startup, steady-state, shutdown) have different signal characteristics. Stage-specific attention mechanisms could improve detection.

Online deployment: Implementing on embedded hardware with low-latency inference for real-time onboard detection.

Uncertainty quantification: Integrating confidence intervals alongside diagnoses would enable more informed decision-making.
Transfer learning: Domain adaptation across different engine configurations would enhance practical applicability.

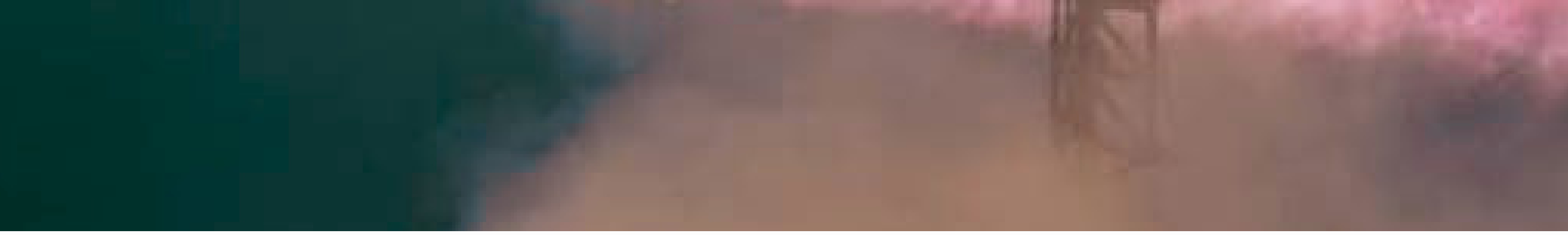


7. CONCLUSION

This paper presented an interpretable fault diagnosis method for liquid rocket engines integrating 1D CNN, Bi-LSTM, and attention mechanism. The approach addresses three critical challenges: accurate diagnosis, model interpretability, and data scarcity.
Key findings:

1. Superior performance: 96% accuracy, outperforming standard CNN and LSTM approaches
2. Inherent interpretability: Attention mechanism provides quantifiable sensor importance rankings and temporal attention distributions
3. Physical alignment: Importance rankings align with LRE operational physics, building trust essential for deployment
4. Data scarcity mitigation: Integration with GAN-based augmentation enables robust training despite limited fault samples

As reusable launch vehicles become central to space exploration, demand for reliable, interpretable health monitoring will intensify. This method represents progress toward AI systems engineers can trust—not because they achieve high accuracy alone, but because they provide actionable insights aligned with physical understanding.



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